Statistics: Standard Deviation

DSAT Math Practice | 25 Questions | Difficulty-Graded | ~15-20% Grid-Ins

Directions: Answer each question by selecting the best answer from the choices given, or by entering your answer in the grid-in box for student-produced response questions. You may use a calculator on all questions. Reference the provided formulas as needed.

1 Easy

Data set A: {3, 3, 3, 3, 3}. Data set B: {1, 2, 3, 4, 5}. Which of the following is true?

A. The standard deviation of A is greater than the standard deviation of B.
B. The standard deviation of A is less than the standard deviation of B.
C. The standard deviations of A and B are equal.
D. There is not enough information to compare the standard deviations.

2 Easy

A data set has a mean of 50 and a standard deviation of 0. Which of the following must be true?

A. The data set contains only one value.
B. Every value in the data set is 50.
C. The data set has an even number of values.
D. The median is greater than the mean.

3 Easy

The ages of five employees at a company are 28, 31, 35, 31, and 40. If the oldest employee (age 40) is replaced by a new employee aged 32, how does the standard deviation change?

A. It increases.
B. It decreases.
C. It stays the same.
D. It cannot be determined.

4 Easy

Which of the following data sets has the greatest standard deviation?

- A {10, 10, 10, 10, 10}
- B {8, 9, 10, 11, 12}
- C {2, 6, 10, 14, 18}
- D {9, 10, 10, 10, 11}

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5 Easy

A teacher records quiz scores for her class: 72, 75, 78, 80, 85. If 5 points are added to every score, which of the following changes?

- A The standard deviation only
- B The mean only
- C Both the mean and the standard deviation
- D Neither the mean nor the standard deviation

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6 Easy

The dot plot shows the number of books read by 10 students last month. Student A read 2 books and Student B read 15 books. Which student's value contributes more to the standard deviation of the data set?

- A Student A, because 2 is a smaller number.
- B Student B, because 15 is farther from the mean.
- C Both contribute equally to the standard deviation.
- D Neither contributes to the standard deviation.

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7 Medium

The mean height of a group of 20 plants is 14 cm with a standard deviation of 3 cm. If each plant grows exactly 2 cm over the next week, what are the new mean and standard deviation?

- A Mean = 16 cm, SD = 3 cm
- B Mean = 16 cm, SD = 5 cm

C Mean = 14 cm, SD = 5 cm

D Mean = 16 cm, SD = 6 cm

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8 Medium

A data set has values {4, 7, 10, 13, 16} with a mean of 10. If every value is doubled, the new standard deviation is how many times the original standard deviation?

A 1/2

B 1

C 2

D 4

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9 Medium

Group X: 12 students with a mean score of 80 and SD = 5. Group Y: 12 students with a mean score of 80 and SD = 12. Which statement must be true?

A Group Y has a higher median.

B Group Y has scores that are more spread out from the mean.

C Group X has a wider range of scores.

D Both groups have the same range.

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10 Medium

A company tracks the daily number of customer complaints over 30 days. The mean is 8 complaints with a standard deviation of 2.5. On one particular day, there were 14 complaints. How many standard deviations above the mean is this value?

A 2

B 2.4

C 3.5

D 5.6

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11 Medium

The data set {5, 8, 10, 12, 15} has a mean of 10. What is the standard deviation of this data set, to the nearest tenth? (Use the population standard deviation formula.)

[Student-Produced Response]

12 Medium

A data set has a mean of 45 and a standard deviation of 6. Which of the following values is within one standard deviation of the mean?

A 33

B 38

C 40

D 52

13 Medium

The table shows test scores for two classes. Class 1: {70, 72, 74, 76, 78}. Class 2: {60, 70, 74, 78, 88}. Which of the following is true?

A Class 1 has a greater mean and greater standard deviation.

B Class 1 has the same mean but a smaller standard deviation.

C Class 2 has a greater mean and greater standard deviation.

D The classes have the same mean but Class 2 has a greater standard deviation.

14 Medium

A researcher collects data on commute times (in minutes) for 50 employees. The mean is 35 minutes with a standard deviation of 8 minutes. A new employee with a commute of 35 minutes joins. What happens to the standard deviation?

A It increases.

B It decreases.

C It stays exactly the same.

D It cannot be determined.

15 Medium

Data set: {20, 24, 24, 28, 32, 36}. The mean is 27.33. If the value 36 is removed, by how much does the mean change? Round to the nearest hundredth.

[Student-Produced Response]

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16 Medium

A data set contains the values 14, 14, 14, 14, 14, and 14. A seventh value of 21 is added. Which of the following must increase?

- A The mean only
- B The standard deviation only
- C Both the mean and the standard deviation
- D Neither the mean nor the standard deviation

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17 Hard

The histogram shows the distribution of exam scores for 100 students. The distribution is approximately symmetric with a mean of 75 and a standard deviation of 8. Approximately what percentage of students scored between 67 and 83?

- A 50%
- B 68%
- C 95%
- D 99.7%

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18 Hard

Data set P has n values with mean M and standard deviation S. If every value in the data set is multiplied by 3 and then 7 is added, what is the new standard deviation?

- A 3S
- B 3S + 7
- C 9S

D 9S + 7

19 Hard

The boxplot shows two data sets, X and Y, both with the same median. Data set X has a much longer box (IQR) than data set Y. Which of the following is most likely true about the standard deviations?

A SD of X < SD of Y

B SD of X = SD of Y

C SD of X > SD of Y

D The relationship cannot be determined from boxplots.

20 Hard

A data set of 6 values has a mean of 12 and a standard deviation of 4. If a seventh value of 12 is added to the data set, what is the new standard deviation, to the nearest tenth? (Use population SD.)

[Student-Produced Response]

21 Hard

Two machines fill bottles. Machine A fills with a mean of 500 mL and SD = 2 mL. Machine B fills with a mean of 500 mL and SD = 8 mL. A quality inspector rejects bottles outside the range 496-504 mL. Which machine will have fewer rejected bottles?

A Machine A

B Machine B

C Both will reject the same number.

D It cannot be determined.

22 Hard

A student's z-score on a math test is -1.5. The class mean is 82 and the standard deviation is 6. What was the student's score?

A 73

B 75

C 79

D 91

23 Hard

The heights (in inches) of 5 basketball players are 72, 74, 76, 78, and 80. What is the population standard deviation of these heights, to the nearest hundredth?

[Student-Produced Response]

24 Hard

A data set has values {a, a, a, b, b} where a < b. The mean of the data set is 20 and the standard deviation is 6. Which system of equations can be used to find a and b?

A $3a + 2b = 100$ and $(3(a-20)^2 + 2(b-20)^2)/5 = 36$

B $3a + 2b = 100$ and $(3(a-20)^2 + 2(b-20)^2)/5 = 6$

C $3a + 2b = 20$ and $(3(a-20)^2 + 2(b-20)^2)/5 = 36$

D $a + b = 20$ and $((a-20)^2 + (b-20)^2)/2 = 36$

25 Hard

Consider two data sets. Data set 1: {k, k, k, k, k} for some constant k > 0. Data set 2: {k-2, k-1, k, k+1, k+2}. If the standard deviation of Data set 2 is d, what is the standard deviation of the data set {2(k-2), 2(k-1), 2k, 2(k+1), 2(k+2)}?

A d

B 2d

C 4d

D d + 2k